b

miércoles, 7 de octubre de 2026 19:17

Cinemática Diferencial

$\frac{\Delta(k+1) - \Delta(k)}{\Delta t} =$

Scara 3gdl (G1-17157)

$\frac{d}{dt}\begin{bmatrix} x = L_1C_1 + L_2C_{12} \\ y = L_1S_1 + L_2S_{12} \\ z = -d_3 \end{bmatrix}$

$\dot{x} = -L_1S_1\dot{\theta}_1 - L_2S_{12}(\dot{\theta}_1 + \dot{\theta}_2)$
$\dot{y} = L_1C_1\dot{\theta}_1 + L_2C_{12}(\dot{\theta}_1 + \dot{\theta}_2)$
$\dot{z} = -\dot{d}_3$

$\dot{x} = (-L_1S_1 - L_2S_{12})\dot{\theta}_1 - L_2S_{12}\dot{\theta}_2$
$\dot{y} = (L_1C_1 + L_2C_{12})\dot{\theta}_1 + L_2C_{12}\dot{\theta}_2$
$\dot{z} = -\dot{d}_3$

$\boxed{\dot{p} = J\dot{q}}$; $\dot{q} = \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{d}_3 \end{bmatrix}$; $\dot{p} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix}$

Recordando
$S_1 = \sin(\theta_1)$.
$C_{12} = \cos(\theta_1 + \theta_2)$.

A (x_A, y_A, z_A) J_A
B (x_B, y_B, z_B) J_B

$$J = \begin{pmatrix} -L_2\sin(\theta_1+\theta_2) - L_1\sin(\theta_1) & -L_2\sin(\theta_1+\theta_2) & 0 \\ L_2\cos(\theta_1+\theta_2) + L_1\cos(\theta_1) & L_2\cos(\theta_1+\theta_2) & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} -L_1S_1 - L_2S_{12} & -L_2S_{12} & 0 \\ L_1C_1 + L_2C_{12} & L_2C_{12} & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{d}_3 \end{bmatrix}$

$\dot{p}$ — J = Jacobiano. $(n \times m)$ — $\dot{q}$

Igual.

$x = L_1C_1 + L_2C_{12}$
$= -L_1S_1 - L_2S_{12} \quad -L_2S_{12} \quad 0$

Método de la cadena

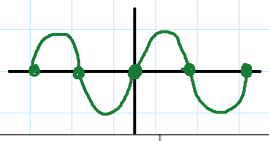
$x = f(\theta_1, \theta_2, d_3)$
$\frac{dx}{dt} = \frac{d}{dt} f(\theta_1, \theta_1, d_3) = \frac{\partial f}{\partial \theta} \cdot \frac{\partial \theta}{\partial t}$

$\dot{p} = J\dot{q}$
$\dot{q} = J^{-1} \cdot \dot{p}$; $J^{-1} = \frac{adj(J)^T}{|J| = 0}$
$\dot{q} = \infty$

Encontrar singularidades

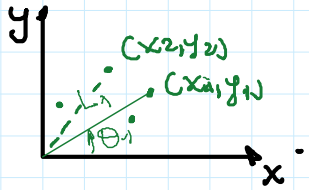
$|J| = 0 \rightarrow |J| = -L_1L_2\sin\theta_2 = 0$
$\theta_2 = 0, \pm\pi$; $n = 1, 2, 3, \ldots \infty$
$\theta_2 = 0^\circ$; $-135^\circ \leq \theta_2 \leq 135^\circ$
$\theta_2 = 0^\circ$.

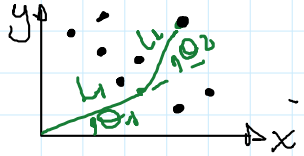
$(L_1, L_2) \neq 0$



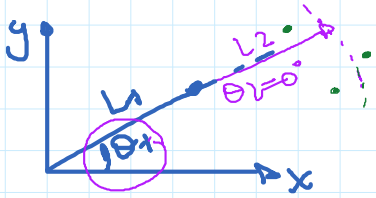
Joints #2 (Cleanroom model)	± 140 deg. (± 140 deg.)	± 152 deg. (± 149 deg.)	± 135 deg. (± 123 deg.)

$\mathbb{R}^2$ (x, y) } 2gdl.





Singularidad $\theta_2 = 0$



Scara 4gdl [TB-4016]

$x = L_1C_1 + L_2C_{12}$
$y = L_1S_1 + L_2S_{12}$
$z = -d_3$
$\psi = \theta_1 + \theta_2 + \theta_4$.

$J_L \begin{cases} x, y, z \\ \theta = [\quad] \end{cases}$

$z = -d_3$

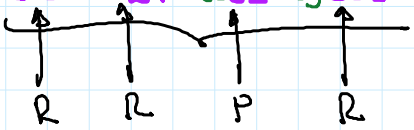
$\psi = \theta_1 + \theta_2 + \theta_4$.

$J = \begin{bmatrix} J_L \\ J_\omega \end{bmatrix}$

$J_L \begin{cases} x, y, z \\ \theta = [\theta_1, \theta_2, d_3, \theta_4] \end{cases}$

$J_\omega = [\varepsilon_0 z_0 \quad \varepsilon_1 z_1 \quad \varepsilon_2 z_2 \quad \varepsilon_3 z_3] = [z_0 \quad z_1 \quad \cdots \quad z_{n-1}]$.

$\varepsilon_i = \begin{cases} 1 & \text{Art. Rotat.} \\ 0 & \text{Art. Prism.} \end{cases}$



$J_\omega = [z_0 \quad z_1 \quad \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad z_3]$: $z_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$z_1 = MO1\,(1:3,3)$

$z_3 = MO3\,(1:3,3)$

$\dot{x} = 2x + 4y$

$\dot{y} = 3x + 1y$

$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{bmatrix} 2 & 4 \\ 3 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} x + \begin{bmatrix} 4 \\ 1 \end{bmatrix} y$